

CNN on Rendered Sample Point Clouds vs. ELA+RF for Black-Box Landscape Classification

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Abstract

Classifying black-box optimization landscapes is important for automated algorithm selection, where the goal is to match a problem instance to a well-suited optimizer using only sampled evaluations. The standard approach, Exploratory Landscape Analysis (ELA), computes aggregate statistical features from a set of sample points, but in doing so discards the spatial and relational structure between those points. We investigate whether a convolutional neural network (CNN) trained on rendered images of sample point clouds can capture this discarded structure and achieve competitive classification accuracy. We render 500 sample evaluations from each of the 24 BBOB benchmark functions in 5 dimensions as color-coded scatter images in two formats: a PCA projection (Type A) and a pairwise dimension grid (Type B). Both CNN models are evaluated under leave-one-problem-out cross-validation against an ELA+Random Forest baseline trained on the same sample budget. The CNN models achieved 0.408 and 0.442 accuracy respectively, compared to 0.650 for ELA+RF. Grad-CAM analysis revealed that the CNN does not consistently attend to the informative regions of the point cloud images, and a sample budget sweep showed that neither method improves beyond 200 sample points. All three methods exhibited high variance across folds, reflecting the difficulty of generalizing across only 24 function classes with 10 instances each. Our results suggest that naively rendering sample point clouds as images for a pretrained CNN does not produce representations expressive enough to compete with ELA on small benchmarks, and that more specialized architectures or larger, more diverse benchmark suites may be needed to make image-based landscape classification viable.

Keywords: Black-box optimization, Exploratory Landscape Analysis, Convolutional neural network, Algorithm selection, BBOB benchmark, Sample point cloud rendering

1 Introduction

Automated algorithm selection has become a growing area of interest in optimization research, as the performance of any given algorithm varies significantly depending on the problem instance it is applied to [1]. In black-box optimization, the fitness function is completely opaque: we cannot inspect its formula, compute gradients, or see its true landscape. The only information available comes from querying the function at chosen input points and observing the scalar output $f(\mathbf{x})$. To put it simply, it is like inserting a stick into the ocean at a number of different points and measuring the depth at each one, without ever seeing the full shape of the ocean floor. Given this constraint, it is important to capture general characteristics of the function landscape from a limited budget of sample points, so that we can determine which algorithms are likely to perform well on functions with similar characteristics.

The standard approach for this is Exploratory Landscape Analysis (ELA) [4], where a set of sample points is used to compute a vector of hand-crafted numerical features describing properties such as y-distribution, meta-model fit, information content, dispersion, and nearest-better clustering [2]. These feature vectors are then fed into a machine learning model to predict algorithm performance or to select the best algorithm for a given problem instance [1]. ELA has proven effective at generating statistical summaries that provide useful input to classifiers and regression models. However, because these features are computed as aggregate statistics over the sample set, they inherently discard the spatial and relational structure between sample points [5]. Two problem instances could produce very similar ELA feature vectors while differing meaningfully in local spatial patterns that influence algorithm behavior. Nikolikj et al. demonstrated this directly: they identified cases where high cosine similarity in ELA feature space did not correspond to similar algorithm performance, indicating that the features were not expressive enough to distinguish between functionally different landscapes [1]. In this project, we explore whether a convolutional neural network (CNN) trained on visual renderings of sample point clouds can learn from the spatial information that ELA discards, and whether this translates to competitive or improved performance in a problem classification task on the BBOB benchmark.

2 Literature Review

Nikolikj, Doerr, and Eftimov [1] studied algorithm performance prediction in a leave-one-problem-out (LOPO) setting, where all instances of a given problem class are removed from training and the model must predict performance on that unseen class. They proposed RF+clust, an approach that calibrates standard Random Forest regression predictions with a weighted average of actual algorithm performance values from the most similar training instances, where similarity is measured by cosine distance in ELA feature space. They evaluated this on three Differential Evolution configurations across the CEC 2014 (30 problems, single instance each) and BBOB (24 problems, five instances each) benchmark suites in dimension 10. Their results showed that RF+clust improved predictions for problems whose ELA representation was close to other training problems where the algorithm also behaved similarly. However, they

also found cases where problems with high feature-space similarity had very different algorithm performance, meaning that the calibration step actually increased prediction error. This finding highlighted a fundamental limitation: ELA features can fail to distinguish between landscapes that look statistically similar but behave differently under optimization, which points toward the need for more expressive problem representations [1].

Kostovska et al. [2] took a different angle by investigating which specific ELA features contribute most to predicting the performance of modular CMA-ES variants. Using the 24 BBOB functions in 5 and 30 dimensions, they trained RF regression models for 40 different CMA-ES configurations across five evaluation budgets (500 to 50,000 function evaluations), and computed SHAP values to quantify each feature’s contribution to prediction accuracy. Their analysis revealed that the set of top-ranked features remained largely the same regardless of the algorithm’s module configuration, suggesting that feature selection does not need to be personalized to each algorithm variant. However, the magnitude of each feature’s influence did vary depending on the configuration. They demonstrated this by training binary classifiers on the SHAP value vectors, which could predict whether a given module (such as elitism or step-size adaptation) was active with accuracy well above the 50% baseline: 79% for elitism in 5D and 76% for step-size adaptation in 5D. This work provides a reference point for which landscape properties carry predictive signal, and our CNN-based approach should, in principle, be able to recover these patterns from the rendered images without being told which features to compute [2].

Cenikj, Petelin, and Eftimov [3] introduced ClustOpt, a methodology for representing and visualizing the search dynamics of optimization algorithms by clustering the solution candidates explored during the optimization process and tracking how cluster memberships evolve across iterations. They applied this to 10 algorithms from the MEALPY library on the BBOB benchmark in 2, 5, and 10 dimensions, and defined two metrics: stability (consistency of an algorithm’s trajectory across different random initializations) and similarity (resemblance of search trajectories between different algorithms). Their results confirmed that variants of the same algorithm family (e.g., DE variants, AEO variants) tended to cluster together in terms of trajectory similarity, validating that the representation captures meaningful behavioral patterns. While ClustOpt focuses on encoding algorithm behavior rather than problem landscapes, it shares a key methodological idea with our project: that structured visual or numerical representations of raw optimization data can serve as informative inputs for machine learning, bypassing the need for hand-crafted feature engineering [3].

3 Problem and Proposed Method

While ELA receives significant attention from the research community, there is a gap in approaches that preserve the spatial, relational, and geometric structure between sample points. ELA features summarize a sample set into aggregate statistics, and in doing so, they throw away information about where points sit relative to each other, how fitness values change across local neighborhoods, and what spatial patterns emerge in the point cloud. Filling this gap with a method that can make use of these

structural cues while still operating within the constraints of black-box optimization is the problem we address in this work.

Our approach replaces ELA feature extraction with a CNN that operates directly on a visual rendering of the sample point cloud. The input to the network is the same set of $N=500$ sample evaluations that would be used for ELA: $(\mathbf{x}, f(\mathbf{x}))$ pairs obtained from black-box queries, where $\mathbf{x}$ is a point in the search space and $f(\mathbf{x})$ is the scalar fitness value returned by the function. Instead of feeding these pairs into statistical feature computation, we render them as a scatter image. Each sample point is plotted in a 2D projection of the input space (obtained via PCA on the input coordinates), and its color encodes the observed fitness value using a viridis colormap applied to rank-based normalized $f(\mathbf{x})$ values. The resulting image is a 128×128 pixel PNG that represents the spatial distribution and fitness structure of the sample cloud in a single visual snapshot. We refer to this as a Type A (PCA scatter) image.

We also generate a second image format, Type B (pairwise grid), which plots scatter panels for pairs of original input dimensions rather than projecting through PCA. This preserves axis-aligned structure that PCA may distort, and is rendered as a 256×192 pixel image with panels arranged in a grid. Both image types are constructed entirely from the sampled query points and their observed outputs, with no access to function formulas, true optima, or gradient information, so the black-box constraint is fully respected.

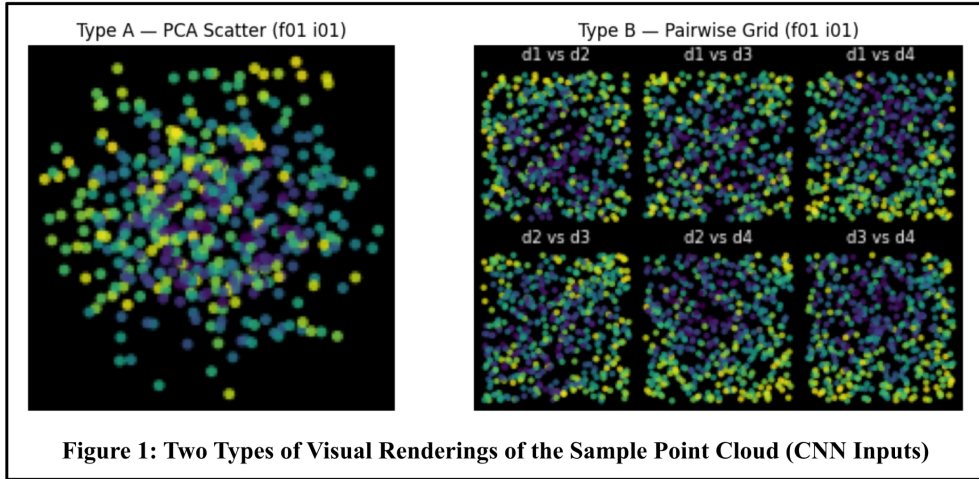


Figure 1: Two Types of Visual Renderings of the Sample Point Cloud (CNN Inputs)

The directness of the representation distinguishes our work from existing approaches. In the standard approach, a practitioner samples the landscape, computes ELA features from those samples, and feeds the feature vector to a classifier. Each step involves design choices and information loss: which features to compute, how to handle feature correlations, what sample size is sufficient for stable feature estimates. Our method removes the middle step entirely. The CNN receives the raw spatial layout of the evidence and learns its own internal features through training. Local density

clusters, color gradients, or texture patterns in the image can carry predictive signals. Whether the CNN can in practice learn to exploit these patterns from limited training data is the central question of this study.

For our baseline comparison, we compute standard ELA features using pflacco from the same $N=500$ sample points and train a Random Forest classifier on the resulting feature vectors, following the approach used by Nikolikj et al. [1] and Kostovska et al. [2]. Both the CNN and the ELA+RF baseline are evaluated under leave-one-problem-out (LOPO) cross-validation on the 24 BBOB problem classes in 5 dimensions, with 10 instances per class. We report accuracy and macro F1 score, and compare performance in a three-way table: CNN (Type A) vs. CNN (Type B) vs. ELA+RF. This controlled comparison, using identical sample budgets and evaluation protocols, allows us to isolate the effect of the representation itself on classification performance.

4 Results

Before reporting results, we note that the BBOB dataset used in this study is both small and unbalanced. The standard three-class grouping of the 24 BBOB functions we adopted is: separable/unimodal (functions 1 through 5), low/moderate conditioning (functions 6 through 14), and multimodal (functions 15 through 24). This gives 5, 9, and 10 functions per class respectively, with 10 instances each, for a total of 240 data points. The imbalance is not just about sample count. Separable/unimodal is represented by fewer distinct function types, so it has less internal variety compared to the other two classes. We chose to accept the dataset as-is rather than subsample the larger classes, since reducing instances would further limit the amount of training data available in an already small dataset, and artificially increasing instances for the underrepresented class would risk overfitting on a narrow set of function types. Since the ELA baseline is trained on the same unbalanced data, the comparison between methods remains fair. A naive majority-class classifier that always predicts multimodal would achieve $10/24 \approx 0.417$ accuracy under LOPO (scoring 1.0 on each of the 10 multimodal folds and 0.0 on all others), which serves as our baseline reference.

To address the class imbalance at the model level, we applied class weights to the cross-entropy loss function in both CNN models. We also replaced the from-scratch CNN architecture with a frozen ResNet-18 backbone (pretrained on ImageNet), training only the final classification head to reduce the number of learnable parameters given the small dataset. On the augmentation side, we applied random rotations, random affine transforms, and Gaussian noise in addition to horizontal flips and color jitter. Finally, we reduced CNN capacity to 2 convolutional layers rather than 4, with smaller dense layer sizes, since a smaller model tends to generalize better with approximately 200 training images. The results reported below reflect these choices already applied.

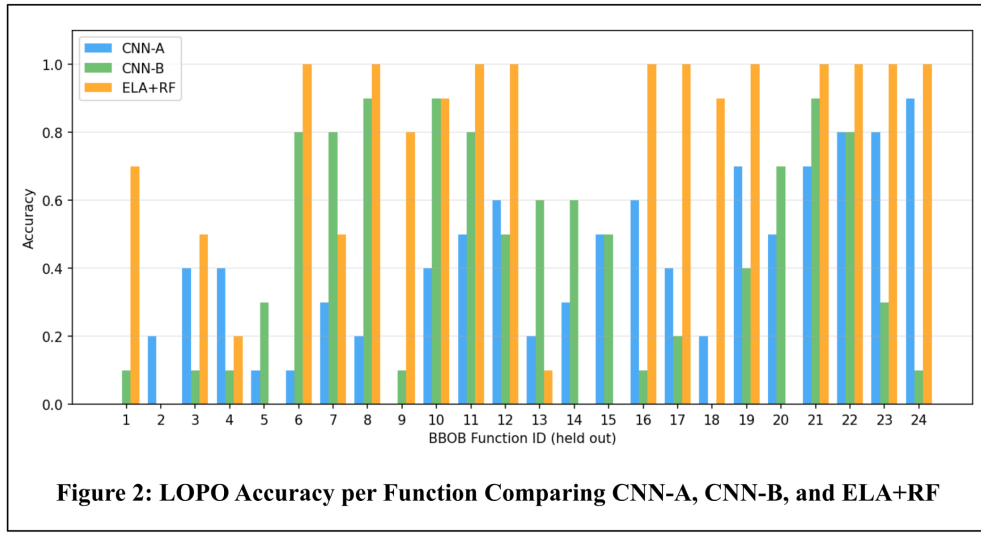
Table 1 summarizes the overall LOPO accuracy and macro F1 score across all 24 folds for each method.

ELA+RF achieved 0.650 overall LOPO accuracy with a standard deviation of 0.425. While the accuracy itself is a reasonable result, the standard deviation is very

Table 1 LOPO classification results across all 24 BBOB folds.

Method	Accuracy	Macro F1
CNN (Type A, PCA)	0.408 ± 0.259	0.200 ± 0.125
CNN (Type B, Pairwise)	0.442 ± 0.327	0.249 ± 0.167
ELA + RF	0.650 ± 0.425	0.559 ± 0.439

large. As visible in Figure 2, ELA+RF scored either 1.0 or 0.0 on 16 of the 24 held-out functions. This all-or-nothing pattern is consistent with the LOPO setting: since each fold holds out 10 instances of the same function, accuracy for that fold can only take values in 0.1 increments (each correct prediction adds 1/10), and ELA either classifies the entire function correctly or fails on it almost completely. The high standard deviation reflects this binary behavior rather than random noise.



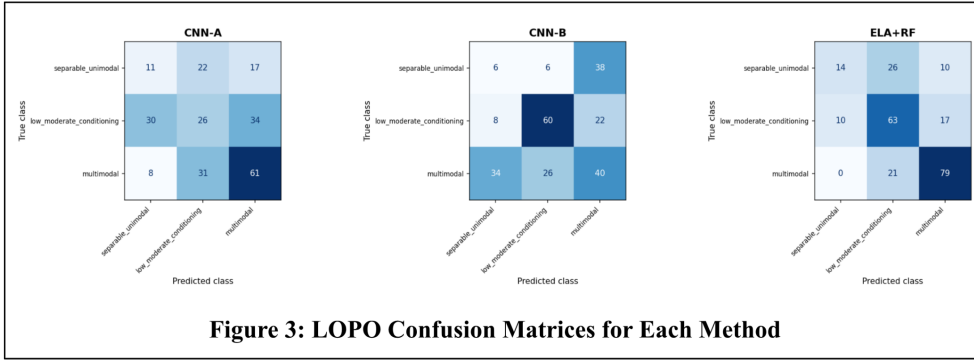
CNN on Type A (PCA) images achieved 0.408 accuracy with 0.259 standard deviation. This is below the majority-class baseline of 0.417, meaning the model performs worse on average than simply predicting the largest class for every input. The macro F1 score of 0.200 confirms that performance is poor across all three classes, not just the underrepresented one. This indicates that a single PCA projection onto the first two principal components does not preserve enough structure for the CNN to learn meaningful class distinctions.

CNN on Type B (Pairwise) images achieved 0.442 accuracy with 0.327 standard deviation. This is a small improvement over Type A and narrowly above the majority-class baseline. The pairwise image format, which plots scatter panels for pairs of original input dimensions rather than projecting through PCA, appears to retain more

of the dimensional structure. This suggests that collapsing the input space into a single 2D PCA projection loses a meaningful amount of information, even for 5-dimensional problems. Still, an accuracy of 0.442 with a standard deviation of 0.327 is far from a reliable classifier.

Figure 2 shows the per-function LOPO accuracy for all three methods side by side. ELA+RF dominates on the majority of functions, particularly in the multimodal class (functions 16 through 24), where it achieves perfect or near-perfect accuracy on most folds. CNN-B shows competitive or better accuracy on a handful of individual functions (such as functions 6, 7, 8, 10, 11, and 21), but is inconsistent across the full set. CNN-A is consistently the weakest performer, rarely exceeding 0.5 on any individual function outside the multimodal class.

The confusion matrices in Figure 3 reveal a more detailed failure pattern. For ELA+RF, the diagonal values are the strongest in each row: 14/50 for separable/unimodal, 63/90 for low/moderate conditioning, and 79/100 for multimodal. While it still struggles with the separable/unimodal class (confusing 26 of 50 instances with low/moderate conditioning), its predictions generally follow the correct trend.



The CNN models show a different and more concerning pattern. CNN-A’s confusion matrix is relatively diffuse, with predictions spread across all three classes without a strong diagonal. For the low/moderate conditioning class, CNN-A predicted separable/unimodal (30) more often than the correct label (26), indicating systematic misclassification rather than random guessing. CNN-B’s confusion matrix is particularly notable: it shows a strong anti-diagonal between separable/unimodal and multimodal. Of the 50 separable/unimodal instances, 38 were predicted as multimodal. Conversely, of the 100 multimodal instances, 34 were predicted as separable/unimodal. The CNN is systematically confusing the two classes that are, by definition, the most different from each other. This reversed pattern suggests that the visual features the CNN is learning from the pairwise scatter images do not correspond to the landscape properties that define these function groups. CNN-B does perform reasonably on the low/moderate conditioning class (60 out of 90 correct), but this is offset by its failure to distinguish the other two.

To better understand why the CNN fails, we applied Grad-CAM to the Type A model to visualize which regions of the input image the network attends to when making predictions. Figure 4 shows three examples: f1 (Sphere, separable/unimodal), f10 (Ellipsoidal rotated, low/moderate conditioning), and f21 (Gallagher 101 peaks, multimodal). The model predicted multimodal, separable/unimodal, and multimodal respectively, getting only the last one correct. In the two incorrect predictions, the hot (red) activation zones in the Grad-CAM overlay fall far from where most sample points cluster, meaning the CNN is basing its decision on sparse, uninformative regions of the image. In the correct prediction for f21, the activation is centered on the dense cluster of sample points. This suggests that the CNN has not learned a consistent strategy for reading the point cloud structure: when it happens to attend to the right region, it can predict correctly, but it does not reliably focus on the informative parts of the image across different functions.

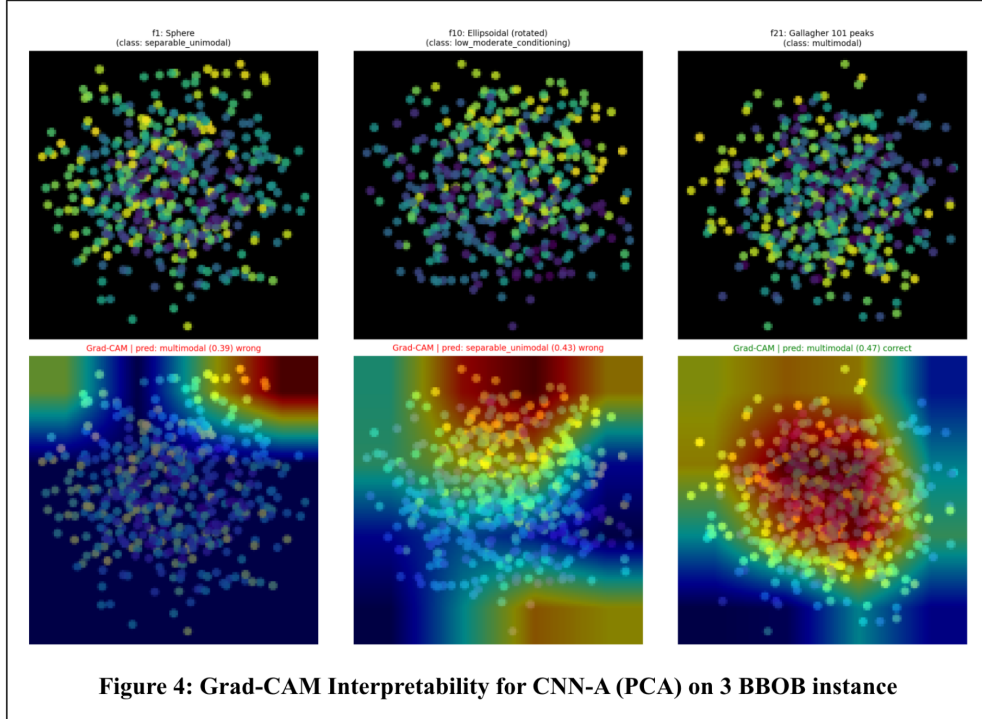
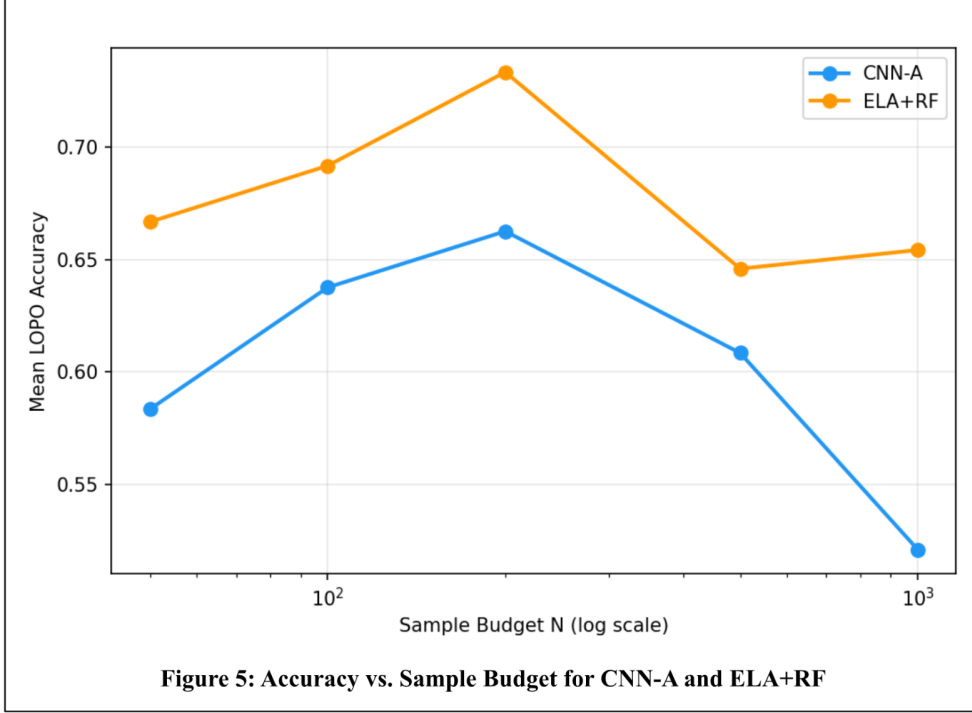


Figure 5 shows how LOPO accuracy changes as a function of the sample budget N for both ELA+RF and CNN-A. Both methods peak at $N=200$, with ELA+RF reaching 0.733 and CNN-A reaching 0.663, and both decline for larger budgets of $N=500$ and $N=1000$. We are not confident in a single explanation for this behavior. One possibility is that denser point clouds produce more cluttered images that are harder for the CNN to parse, and that ELA feature estimates become noisier or more correlated as the sample set grows beyond what the statistical estimators were designed

for. Regardless of the cause, the fact that performance does not improve monotonically with more data suggests that both representations have a capacity ceiling that is not addressed simply by increasing the sample budget.



Overall, our image-based CNN models did not outperform ELA+RF in BBOB function type classification. CNN-A fell below the majority-class baseline, and CNN-B only narrowly exceeded it. ELA+RF, despite its own high variance, achieved meaningfully better accuracy and macro F1 scores across all classes. This suggests that, in the environment we tested, the CNN does not learn from the rendered sample point clouds more effectively than ELA learns from its computed feature vectors. It is worth noting that ELA also showed considerable variance (0.425 standard deviation), which is consistent with the small dataset size and the difficulty of the LOPO setting in general. The limited number of distinct functions per class, especially for separable/unimodal with only 5 functions, makes generalization difficult for any method. Future work could explore whether a larger and more balanced benchmark suite, or a CNN architecture specifically designed for this type of scientific scatter plot input rather than adapted from natural image classification, could close the gap between image-based and feature-based approaches.

AI Disclosure

AI tools (Claude, ChatGPT) were used throughout this project for assistance with writing and debugging code, brainstorming ideas, and refining (grammar and tone) the project paper. AI was not used for producing experimental results or performing analysis. All model training, evaluation, and interpretation of results were done independently.

References

- [1] Nikolikj, A., Doerr, C., Eftimov, T.: RF+clust for Leave-One-Problem-Out Performance Prediction. In: *Applications of Evolutionary Computation (EvoApplications 2023)*, Lecture Notes in Computer Science. Springer (2023). arXiv:2301.09524
- [2] Kostovska, A., Vermetten, D., Džeroski, S., Doerr, C., Korošec, P., Eftimov, T.: The Importance of Landscape Features for Performance Prediction of Modular CMA-ES Variants. In: *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO 2022)*. ACM (2022). arXiv:2204.07431
- [3] Cenikj, G., Petelin, G., Eftimov, T.: ClustOpt: A Clustering-based Approach for Representing and Visualizing the Search Dynamics of Numerical Metaheuristic Optimization Algorithms. In: *Proceedings of the IEEE Congress on Evolutionary Computation (CEC 2025)*. IEEE (2025). arXiv:2507.02337
- [4] Mersmann, O., Bischl, B., Trautmann, H., Preuss, M., Weihs, C., Rudolph, G.: Exploratory Landscape Analysis. In: *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO 2011)*, pp. 829–836. ACM (2011)
- [5] Kerschke, P., Hoos, H.H., Neumann, F., Trautmann, H.: Automated Algorithm Selection: Survey and Perspectives. *Evolutionary Computation* **27**(1), 3–45 (2019)